# Physics P1: Physical Reasoning and Estimation

MAIS Physics & Astronomy Club (PAC)

Before solving differential equations or memorizing long formulas, physicists ask simpler questions: **Do the units match? What happens in extreme cases? Roughly how big should the answer be?** These three habits catch algebraic mistakes immediately and often reveal the form of a physical law before doing a single integral. Problems are marked from **[1]** (warm-up) to **[3]** (challenge).

## 1. Units and Dimensional Analysis

Every physical quantity in mechanics can be built from three base dimensions: length [**L**], mass [**M**], and time [**T**]. Two quantities may be added, subtracted, or equated only if their dimensions match, and the argument of any transcendental function (such as sin **x**, $\mathbf{e^x}$, or ln **x**) must be dimensionless.

**Example 1.** A simple pendulum consists of a small bob of mass **m** hanging on a light string of length $\ell$ in a uniform gravitational field **g**. How does the period **P** of small oscillations depend on **m**, $\ell$, and **g**?

**Solution.** Write $\mathbf{P} = \mathbf{C}\,\mathbf{m}^{\mathbf{a}}\,\ell^{\mathbf{b}}\,\mathbf{g}^{\mathbf{c}}$, where **C** is a dimensionless constant. Comparing dimensions gives

$$[\mathbf{T}] = [\mathbf{M}]^{\mathbf{a}}\,[\mathbf{L}]^{\mathbf{b}}\,([\mathbf{L}][\mathbf{T}]^{-2})^{\mathbf{c}} = [\mathbf{M}]^{\mathbf{a}}\,[\mathbf{L}]^{\mathbf{b}+\mathbf{c}}\,[\mathbf{T}]^{-2\mathbf{c}}. \tag{1}$$

Matching exponents of [**M**], [**T**], and [**L**] yields $\mathbf{a} = 0$, $-2\mathbf{c} = 1 \Rightarrow \mathbf{c} = -1/2$, and $\mathbf{b} + \mathbf{c} = 0 \Rightarrow \mathbf{b} = 1/2$. Therefore

$$\mathbf{P} = \mathbf{C}\,\sqrt{(\ell)/\,(\mathbf{g})}. \tag{2}$$

Newton's laws are only needed to fix the numerical prefactor $\mathbf{C} = 2\pi$. Even without solving the equation of motion, dimensional analysis already tells us that doubling the mass has no effect, while quadrupling the length doubles the period.

**Remark.** Why did we restrict to **small** oscillations? If the release angle $\theta_0$ is large, $\theta_0$ is itself a dimensionless parameter, so **C** becomes a function $\mathbf{f}(\theta_0) = 2\pi(1 + \theta_0^{\,2}/16 + \cdots)$. Always check whether hidden dimensionless ratios exist!

**[1] Problem 1.** Ocean ripples and waves carry energy across water of density $\rho$ under gravitational acceleration $\mathbf{g}$.

**(a)** For waves of wavelength $\lambda$ on very deep water (depth $\mathbf{h} \gg \lambda$), the wave speed $\mathbf{v}$ can depend only on $\lambda$, $\mathbf{g}$, and $\rho$. Use dimensional analysis to find how $\mathbf{v}$ scales with $\lambda$, $\mathbf{g}$, and $\rho$.

**(b)** For tsunami waves in an ocean whose depth $\mathbf{h}$ is much smaller than the wavelength ($\mathbf{h} \ll \lambda$), experiments show that $\mathbf{v}$ is independent of $\lambda$. Find how $\mathbf{v}$ depends on $\mathbf{h}$ and $\mathbf{g}$, and estimate the speed (in km/h) of a tsunami over a 4.0 km deep ocean basin.

## 2. Limiting Cases and Symmetry

Whenever you derive an algebraic expression, test it against cases where you already know what must happen physically: setting an angle to 0 or 90°, letting one mass become much heavier than another ($\mathbf{M} \gg \mathbf{m}$), or swapping two identical labels.

**Example 2.** Two blocks of masses $\mathbf{m}_1$ and $\mathbf{m}_2$ hang on opposite sides of a massless, frictionless pulley (an Atwood machine). Suppose a student writes down $\mathbf{a} = (\mathbf{m}_1 + \mathbf{m}_2)/(\mathbf{m}_1 - \mathbf{m}_2)\,\mathbf{g}$ for the downward acceleration of $\mathbf{m}_1$. How do we see immediately that this is wrong?

**Solution.** If $\mathbf{m}_1 = \mathbf{m}_2$, the system is balanced and must have $\mathbf{a} = 0$, whereas the student's formula diverges to infinity! The correct result,

$$\mathbf{a} = (\mathbf{m}_1 - \mathbf{m}_2)/(\mathbf{m}_1 + \mathbf{m}_2)\,\mathbf{g}, \tag{3}$$

vanishes when $\mathbf{m}_1 = \mathbf{m}_2$, approaches $+\mathbf{g}$ (free fall) when $\mathbf{m}_1 \gg \mathbf{m}_2$, and changes sign when $\mathbf{m}_1$ and $\mathbf{m}_2$ are interchanged.

**[2] Problem 2.** A small block of mass $\mathbf{m}$ is placed on a frictionless wedge of mass $\mathbf{M}$ and inclination angle $\theta$, which itself rests on a frictionless horizontal table. Five candidates are proposed for the magnitude $\mathbf{A}$ of the wedge's horizontal acceleration:

(i) $(\mathbf{mg}\cos\theta)/(\mathbf{M}+\mathbf{m})$, (ii) $(\mathbf{Mg}\sin\theta\cos\theta)/(\mathbf{M}+\mathbf{m}\sin^2\theta)$, (iii) $(\mathbf{mg}\sin\theta\cos\theta)/(\mathbf{M}+\mathbf{m}\sin^2\theta)$, (iv) $(\mathbf{mg}\tan\theta)/(\mathbf{M}+\mathbf{m})$, (v) $(\mathbf{mg}\sin\theta\cos\theta)/(\mathbf{m}+\mathbf{M}\sin^2\theta)$.

**(a)** By examining limiting cases such as $\theta \to 0$, $\theta \to \pi/2$, $\mathbf{M} \to \infty$, and $\mathbf{M} \gg \mathbf{m}$ at small $\theta$ (without solving Newton's equations from scratch), eliminate four of the five expressions and identify the only physically consistent candidate.

**(b)** Verify the surviving formula directly by writing down Newton's second law for the block and the wedge.

## 3. Order-of-Magnitude Estimation

An order-of-magnitude (or Fermi) estimate breaks an unfamiliar quantity into pieces you know within a factor of 2 or 3. Overestimates and underestimates tend to partially cancel when multiplied together, frequently landing remarkably close to the exact value.

**[2] Problem 3.** Sea-level atmospheric pressure is $\mathbf{P}_0 \approx 1.01 \times 10^5$ Pa, Earth's radius is $\mathbf{R}_\oplus \approx 6.37 \times 10^6$ m, and $\mathbf{g} \approx 9.8$ m/s$^2$.

**(a)** Explaining why the atmosphere's thickness (≈ 10 km) being small compared to $\mathbf{R}_\oplus$ simplifies the calculation, estimate the total mass $\mathbf{M}_{\text{atm}}$ of Earth's atmosphere.

**(b)** Ulaanbaatar sits at an elevation of roughly 1350 m above sea level, where typical atmospheric pressure is about $8.6 \times 10^4$ Pa. Using the same idealized vertical-column model, approximately what fraction of a sea-level air column's mass lies **below** the altitude of Ulaanbaatar? (The global fraction below 1350 m also depends on topography.)

**[3] Problem 4 (Collapsing clouds and stars).** Suppose all internal pressure inside a spherical cloud or star of uniform initial density $\rho_0$ suddenly vanished, so that it collapsed inward solely under its own gravity (with Newton's constant $\mathbf{G} = 6.67 \times 10^{-11}$ N·m$^2$/kg$^2$).

**(a)** Using dimensional analysis, find how the collapse time $\mathbf{t}_{\text{ff}}$ depends on $\mathbf{G}$, $\rho_0$, and the initial radius $\mathbf{R}_0$. Explain physically why $\mathbf{R}_0$ drops out when $\rho_0$ is held fixed.

**(b)** Estimate $\mathbf{t}_{\text{ff}}$ in minutes for the Sun (mass $\mathbf{M}_\odot = 1.99 \times 10^{30}$ kg, radius $\mathbf{R}_\odot = 6.96 \times 10^8$ m). Why has the real Sun shone steadily for 4.6 billion years instead of collapsing on this timescale?